

Beautiful Soup Full Notes

1. Introduction

Beautiful Soup is a Python library used for parsing HTML and XML documents and extracting data from web pages.

2. Installation

```
pip install beautifulsoup4
pip install requests
```

3. Basic Example

```
import requests
from bs4 import BeautifulSoup

url = "https://example.com"
response = requests.get(url)

soup = BeautifulSoup(response.text, "html.parser")
print(soup)
```

4. Finding Elements

```
soup.find("h1")
soup.find_all("p")
soup.find("p", class_="text")
soup.find(id="main")
```

5. Extract Text

```
tag = soup.find("h1")
print(tag.text)
```

6. Attributes

```
link = soup.find("a")
print(link["href"])
```

7. Loop Example

```
items = soup.find_all("p")
for item in items:
    print(item.text)
```

8. CSS Selectors

```
soup.select("p.text")
```

```
soup.select("#main")
soup.select("div p")
```

9. Real Scraping Example

```
import requests
from bs4 import BeautifulSoup

url = "https://quotes.toscrape.com"
response = requests.get(url)

soup = BeautifulSoup(response.text, "html.parser")
quotes = soup.find_all("span", class_="text")

for q in quotes:
    print(q.text)
```

10. Save to File

```
with open("data.txt", "w") as f:
    f.write(soup.text)
```

11. Parsers

```
BeautifulSoup(html, "html.parser")
BeautifulSoup(html, "lxml")
BeautifulSoup(html, "html5lib")
```

12. Advanced

```
soup.get_text()
tag.text.strip()
tag.get("href")
```

13. Pagination

```
for i in range(1, 5):
    url = f"https://example.com/page/{i}"
    response = requests.get(url)
    soup = BeautifulSoup(response.text, "html.parser")
    print(soup.title.text)
```

14. Headers

```
headers = {"User-Agent": "Mozilla/5.0"}
response = requests.get(url, headers=headers)
```

15. Limitations

Cannot handle JavaScript rendering. Use Selenium or Playwright for dynamic sites.

16. Best Practices

Respect robots.txt, use delays, and avoid overloading servers.